

1) Simulation ~~3BP~~

EOM in Jacobi coord

dimensionless $m_1 + m_2 + m_3 = 1$

$$G = 1$$

inputs: masses, IC's $(\vec{R}, \dot{\vec{R}})$

outputs: $\vec{R}, \dot{\vec{R}}$ over time, t

2) $m_1 \neq m_2 \neq m_3$ $\vec{R}_0, \dot{\vec{R}}_0$

plot time history & variation

- Routh stability criteria

$$m_1 + m_2 + m_3 = 1$$

$$m_1 m_2 + m_2 m_3 + m_3 m_1 < \frac{1}{27}$$

$$m_1^2 + 2m_1 m_3 < \frac{1}{27}$$

$$2m_1 + m_3 = 1$$

$$m_3 = 1 - 2m_1$$

Computational hwk #1

Nicole James

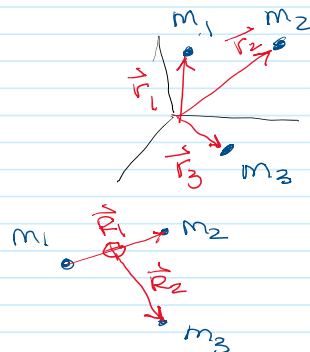
3/20/26

1) given: 3BP EOM in Jacobi coord

$$m_1 + m_2 + m_3 = 1 \quad G = 1$$

input: masses, $\vec{R}_0$, $\vec{V}_0$

output: time series of $\vec{r}$ & $\dot{\vec{r}}$



$$\vec{r}_{12} = \vec{r}_2 - \vec{r}_1$$

$$\vec{r}_{23} = \frac{1}{m_1 + m_2} [m_1(\vec{r}_3 - \vec{r}_1) + m_2(\vec{r}_3 - \vec{r}_2)]$$

$$\vec{r}_N = \vec{r}_c$$

$$\vec{r}_{N-1} = \frac{1}{\sum_{i=1}^{N-1} m_i} \sum_{i=1}^{N-1} m_i \vec{r}_{iN}$$

$$\vec{r}_{12} = \vec{r}_1$$

$$\vec{r}_{23} = \vec{r}_2 - \frac{m_1}{m_1 + m_2} \vec{r}_1$$

$$\vec{r}_{13} = \vec{r}_2 - \frac{m_2}{m_1 + m_2} \vec{r}_1$$

$$M_i = \sum_{j=1}^N m_j$$

$$T = \sum_{i=1}^{N-1} M_{i+1} \frac{M_{i+1} m_i}{M_i} \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i + \frac{1}{2} M_N \dot{\vec{r}}_N \cdot \dot{\vec{r}}_N$$

$$H = \sum_{i=1}^{N-1} \frac{M_{i+1} m_i}{M_i} \dot{\vec{r}}_i \times \dot{\vec{r}}_i + M_N \dot{\vec{r}}_N \times \dot{\vec{r}}_N$$

$$I_p = \sum_{i=1}^{N-1} \frac{M_{i+1} m_i}{M_i} R_i^2 + M_N R_N^2$$

$$\vec{y} = \begin{bmatrix} \vec{r} \\ \dot{\vec{r}} \end{bmatrix} \quad \frac{d\vec{y}}{dt} = \dot{\vec{y}} = \begin{bmatrix} \dot{\vec{r}} \\ \ddot{\vec{r}} \end{bmatrix}$$

$$U = - \sum_{i=1}^{N-1} \sum_{j=i+1}^N \frac{G m_i m_j}{|\vec{r}_{ij}|}$$

Lagrangian Eqs $\Rightarrow$ EOM

$$\frac{\partial \vec{r}_i}{\partial \vec{r}_j} = \begin{cases} 0 & j \geq i \\ 1 & j = i-1 \\ \frac{m_{j+1}}{m_{j+1} + 1} & j \leq i-2 \end{cases}$$

$$\frac{\partial U}{\partial \vec{r}_k} = -G \sum_{l=1}^{N-1} \sum_{j=l+1}^N \frac{m_l m_j}{r_{lj}^3} \vec{r}_{lj} \cdot \frac{\partial \vec{r}_{lj}}{\partial \vec{r}_k}$$

$$L = T - U$$

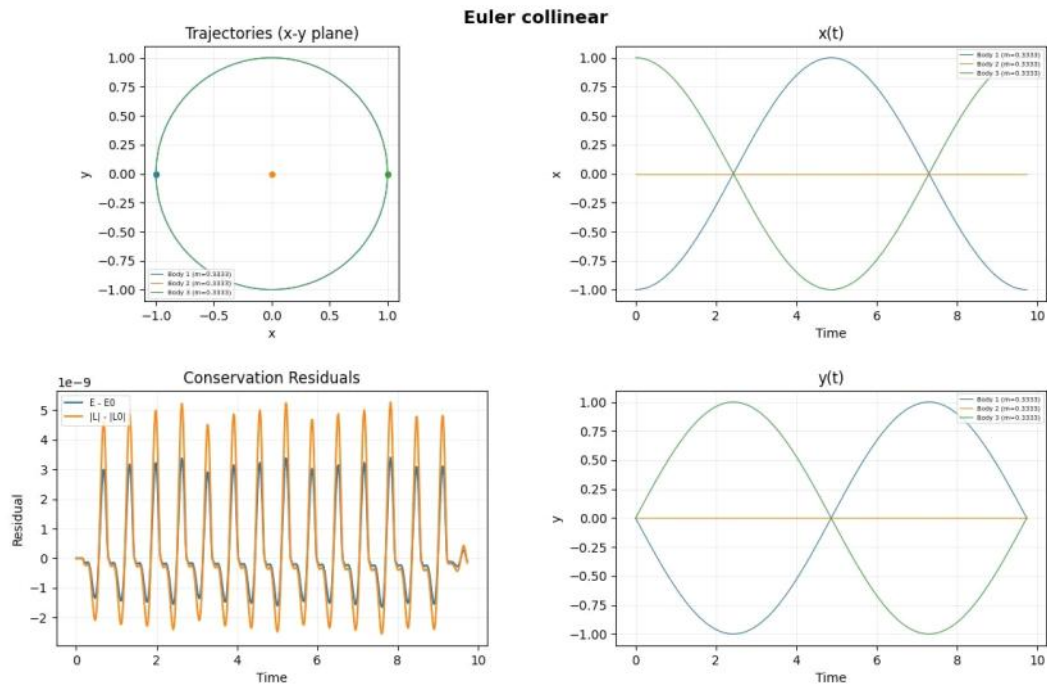
$$\frac{\partial L}{\partial \vec{r}_j} = \frac{M_{j+1} m_j}{M_j} \dot{\vec{r}}_j$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\vec{r}}_j} \right) = \frac{M_{j+1} m_j}{M_j} \ddot{\vec{r}}_j$$

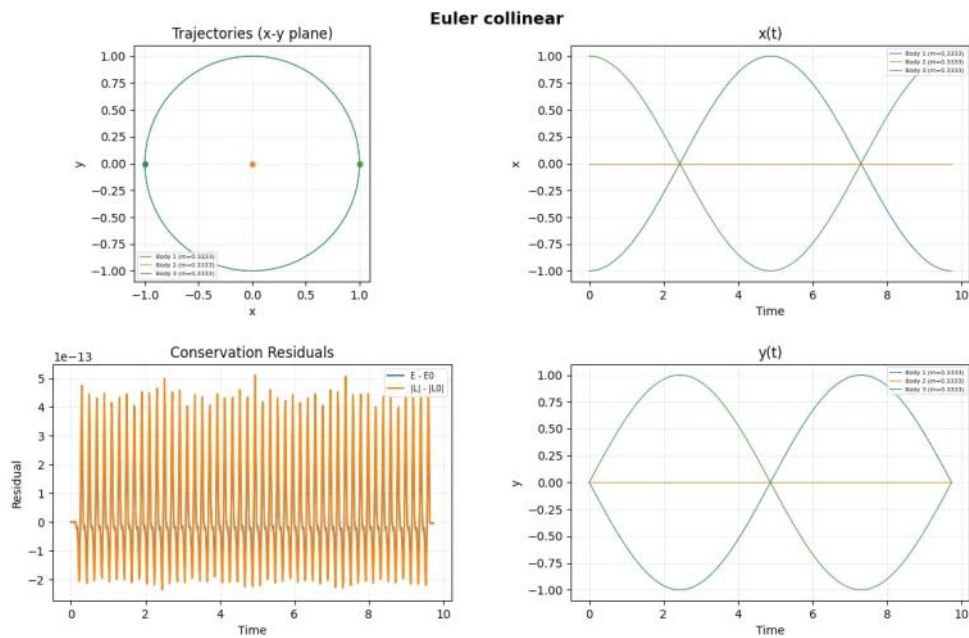
1a) rel tol = 1×10^{-8}
abs tol = 1×10^{-10}

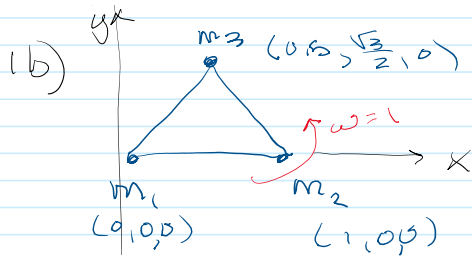
(using Euler
const mass conf(q))

1a) $\text{rel tol} = 1 \times 10^{-8}$ (using Euler
 $\text{abs tol} = 1 \times 10^{-10}$ const mass config)



$\text{rel tol} = 1 \times 10^{-12}$
 $\text{abs tol} = 1 \times 10^{-14}$



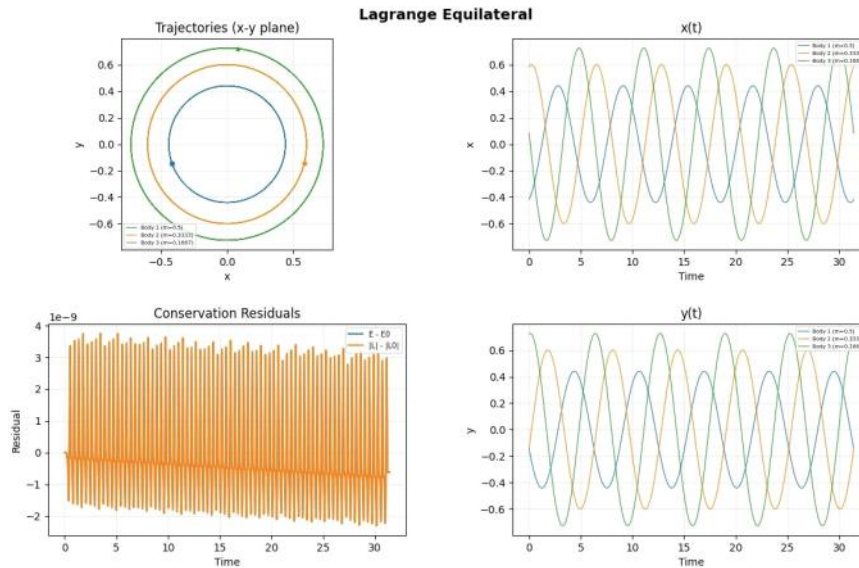


$$m_1 = \frac{1}{2} \quad m_2 = \frac{1}{3} \quad m_3 = \frac{1}{6}$$

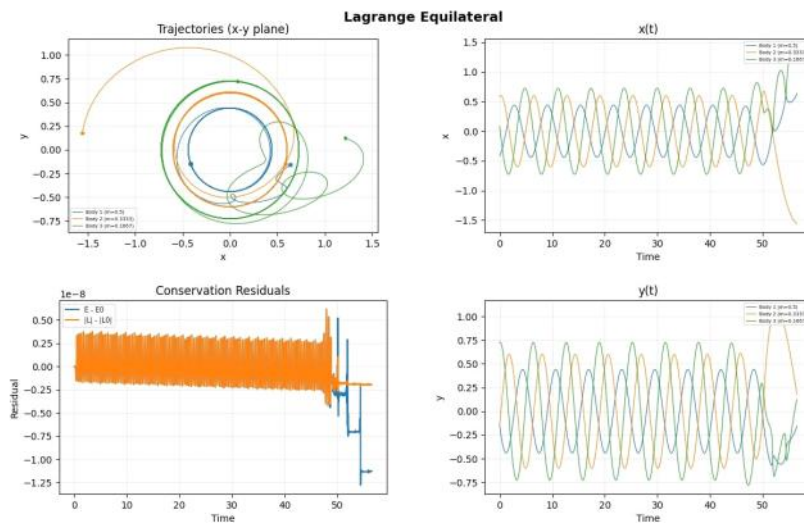
$$\omega = 1 \quad \vec{v} = \omega \vec{r}$$

$$\text{period} = 2\pi / \omega$$

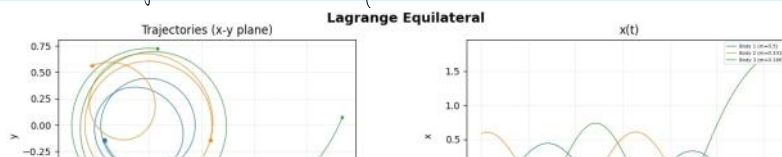
The orbit stays stable through about 8 periods

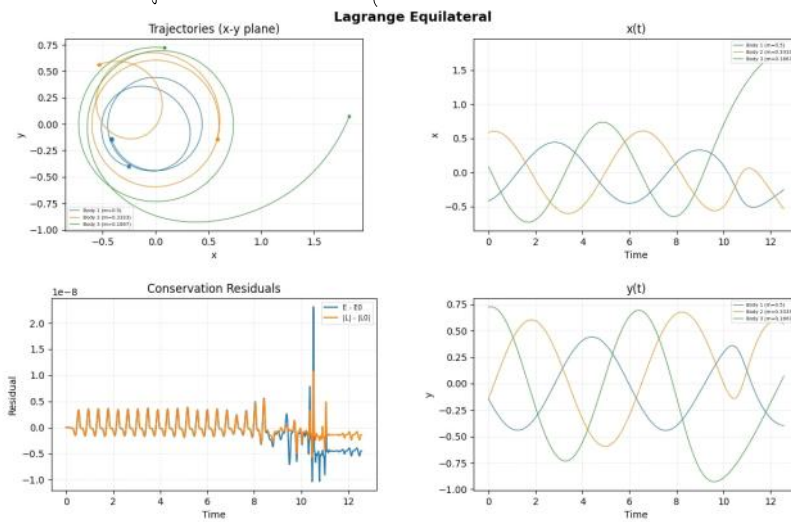


During the 9th period, it becomes quite chaotic

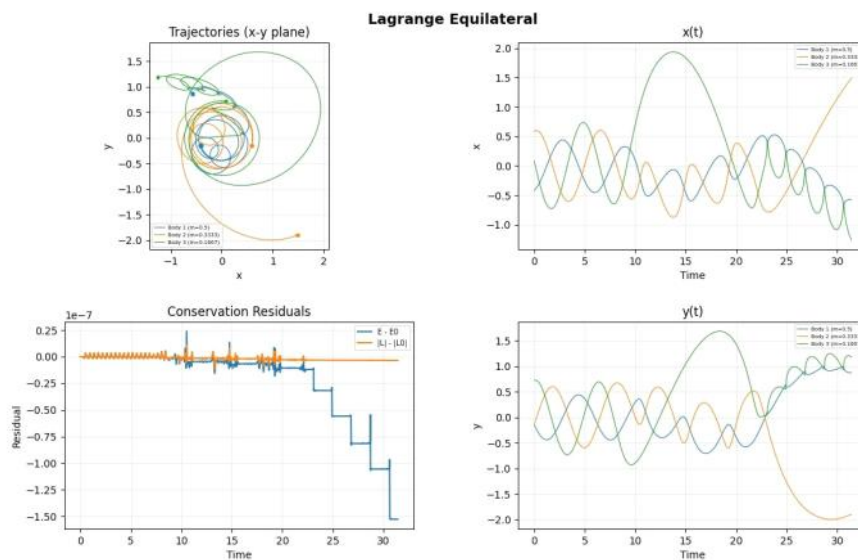


Adding a small perturbation to a single body (body 2) results in a large difference in each body's trajectory, this after 2 periods





This is after 5 periods:



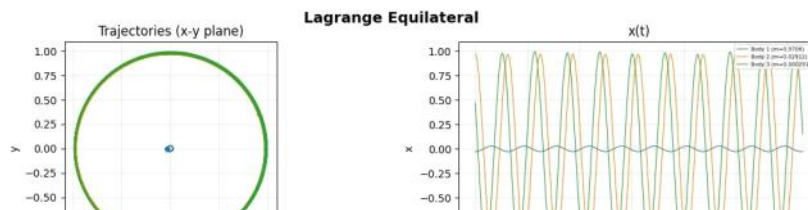
c) repeating (b) but with masses that meet the Routh stability criterion

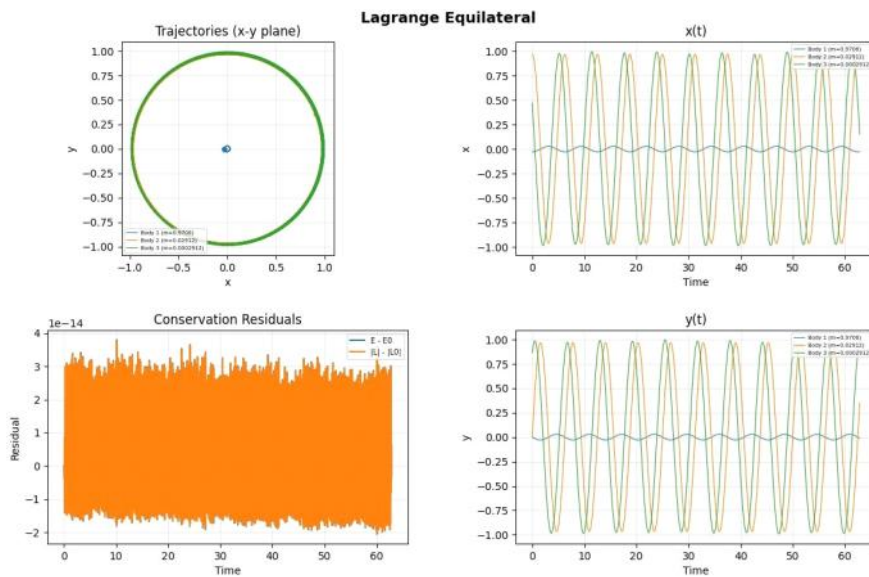
$$m_1 = \frac{1}{m_1 + m_2 + m_3}$$

$$m_2 = \frac{0.03}{m_1 + m_2 + m_3}$$

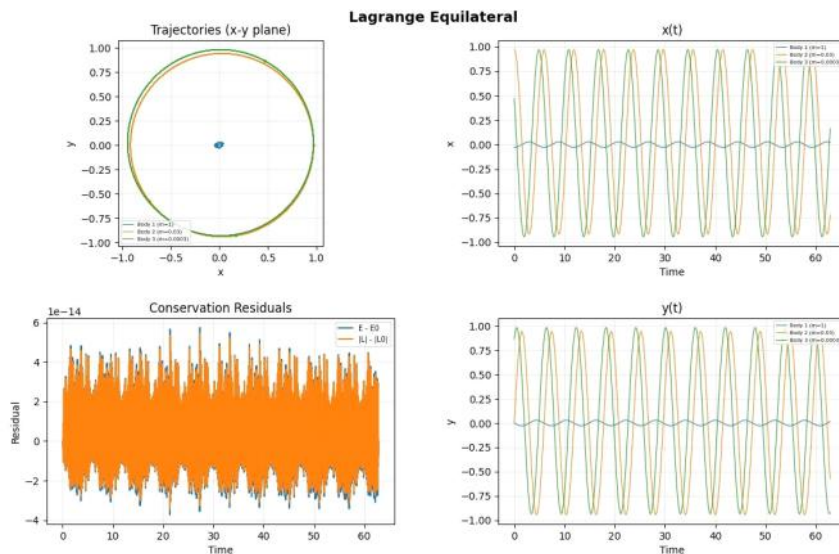
$$m_3 = \frac{0.0003}{m_1 + m_2 + m_3}$$

They are stable & stay in equilateral triangle config



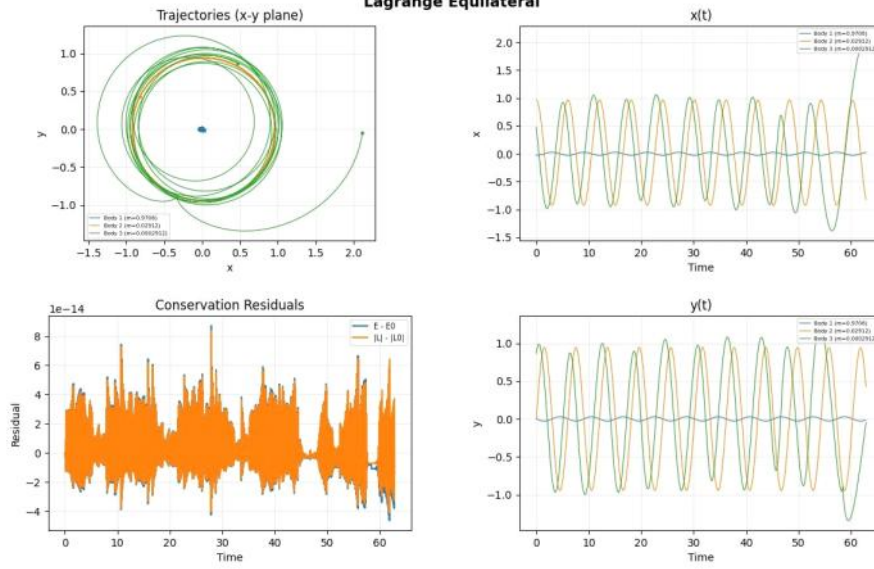


Adding 0.001 to r_2 results in small deviations but system is still stable & in equilateral triangle config.



The system stays stable thru the 1st 10 orbits until I perturb m_2 by 0.007. Then the system config moves away from triangular config.

Lagrange Equilateral



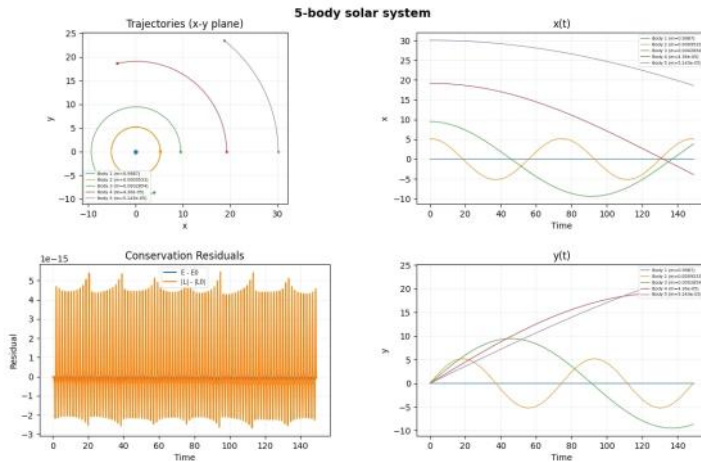
Problem 2

2a) I used the outer planets to verify
SBP - Sun, Jupiter, Saturn, Uranus, Neptune

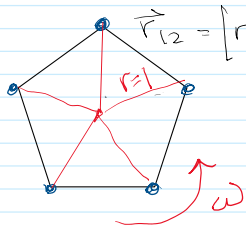
$$m_{\text{Sun}} = 1 \quad m_J = 9.54 \times 10^{-4} \quad m_S = 2.86 \times 10^{-4} \quad m_U = 4.37 \times 10^{-5} \quad m_N = 5.15 \times 10^{-5}$$

$$V = m_{\text{Sun}}/r \quad r_J = 5.2 \text{ Au} \quad r_S = 9.5 \text{ Au} \quad r_U = 19.2 \text{ Au} \quad r_N = 30.1 \text{ Au}$$

Energy & momentum are conserved thru
2 Jupiter orbits.



2b)

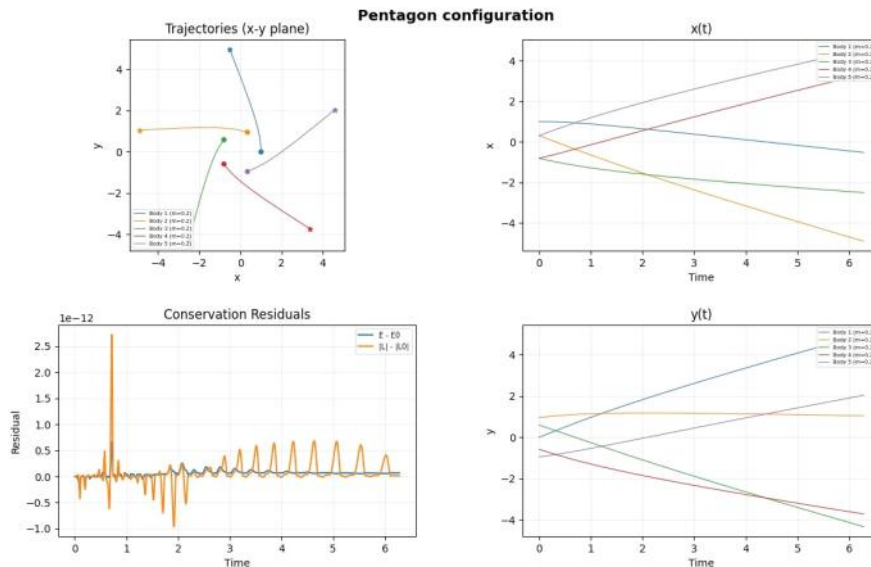


$$\vec{r}_i = [r \cos(2\pi i/5), r \sin(2\pi i/5), 0]$$

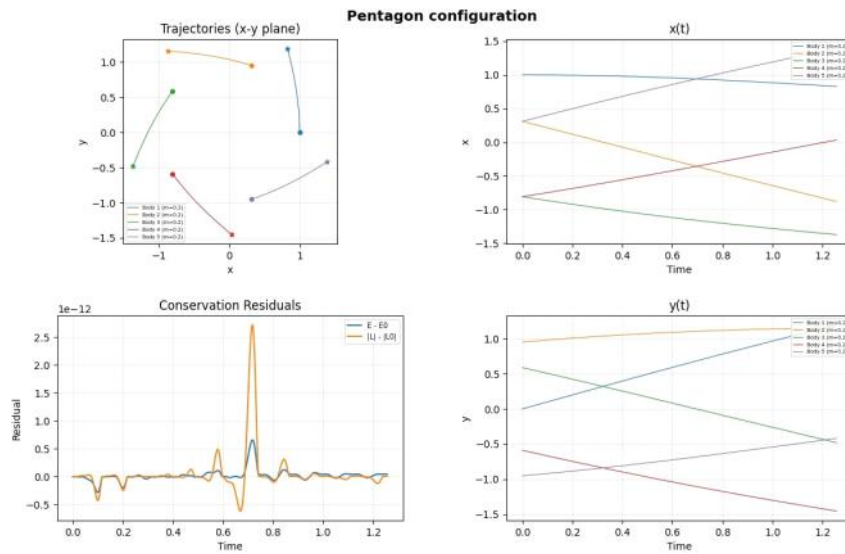
$$m_i = \frac{1}{5} \quad \omega = \frac{GM}{r^3}$$

$$v_i = \omega \vec{r}_i$$

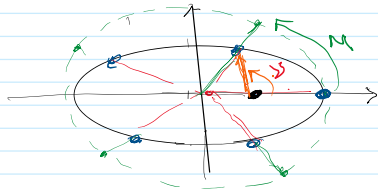
In this configuration the bodies fly
away immediately.
Here is one period:



Here is 0.2 period



2c) 5 bodies in eccentric orbit

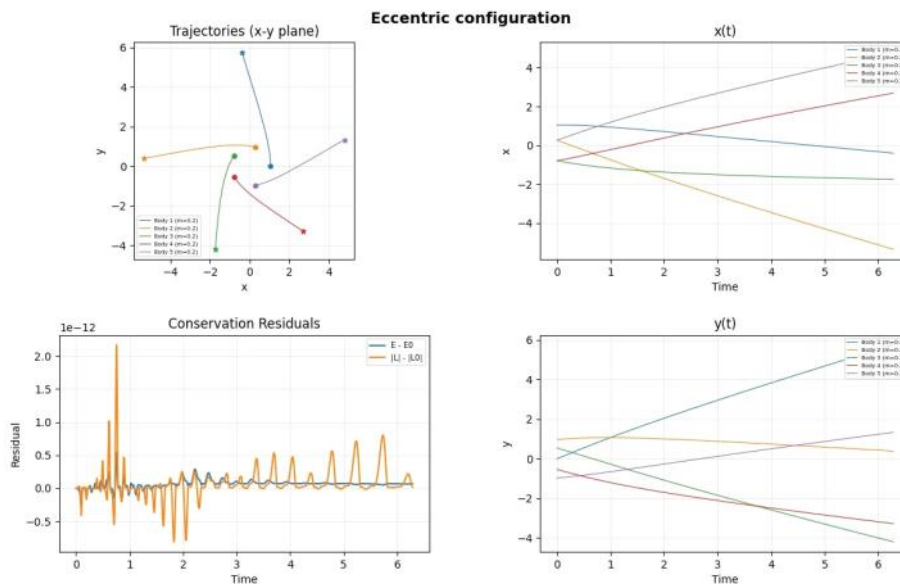


$$a = 1 \quad e = 0.1 \quad \text{Period} = 2\pi \sqrt{\frac{a^3}{\mu}}$$

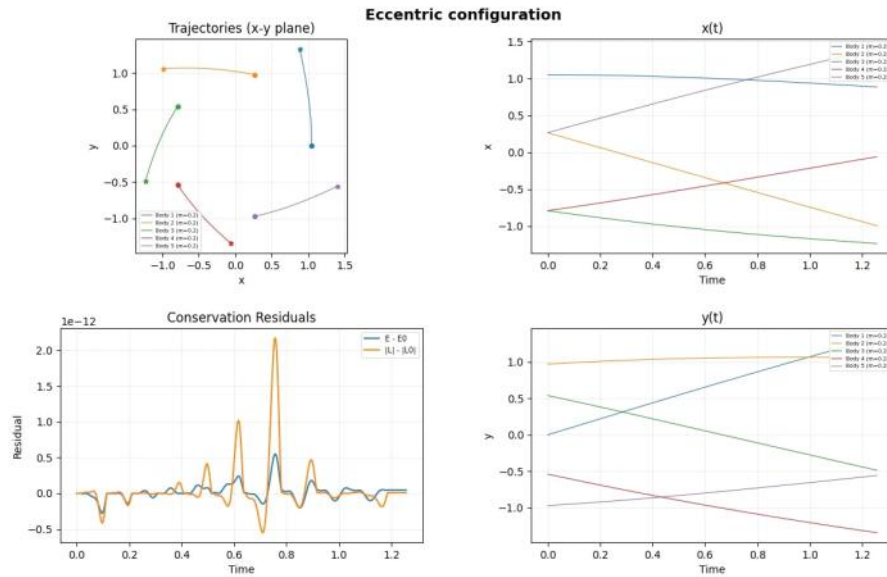
$$h^2 = \mu a (1 - e^2)$$

M = mean anomaly
 $f = E - E_0 \sin(e) - M$
 $\tan \frac{\gamma}{2} = \sqrt{\frac{1+e}{1-e}} \tan \frac{E}{2}$

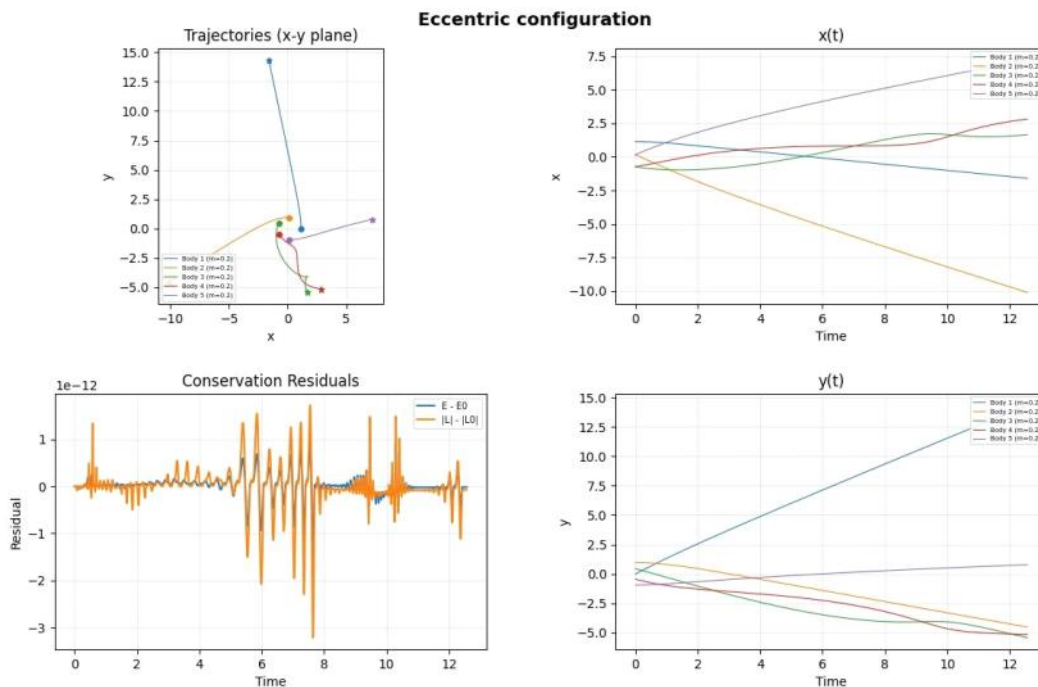
The system is still unstable & the masses fly away from each other immediately but since there the masses have non uniform velocity, the orbit departures are "shallower" since they are on an ellipse the velocity components are slightly different.



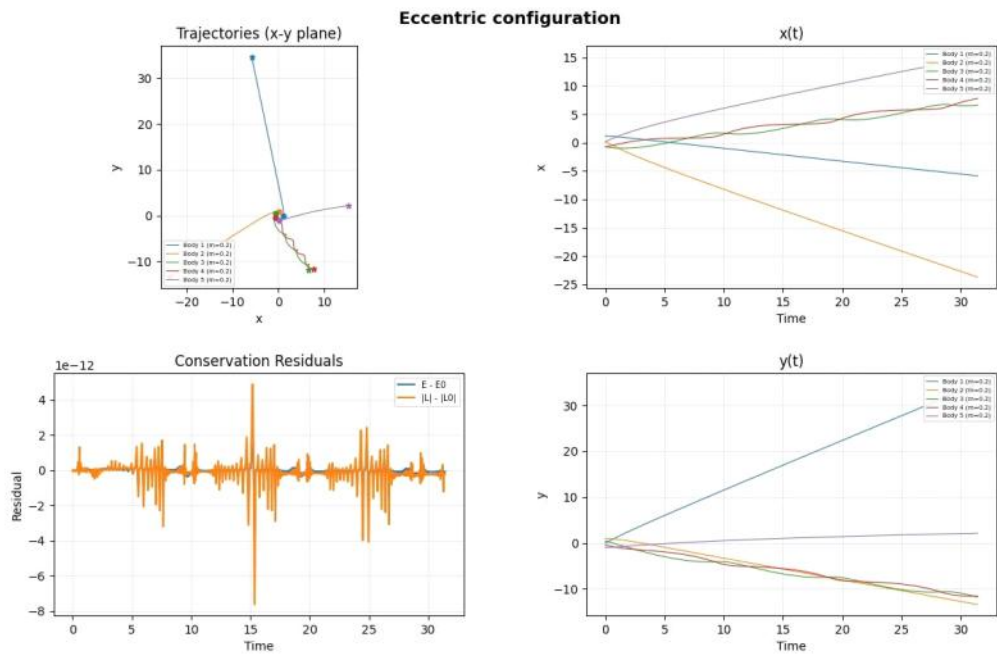
This 0.2 period



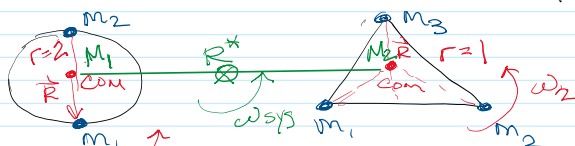
As the eccentricity is increased, the body @ periaapsis leaves the system fastest & travels farther a way than the other bodies.
 This is $e = 0.3$ over the course of 2 periods. The bodies near apoapsis continue to impact the others orbit because their velocity is slowest.



After 5 periods they are still bound to each other.



2d) 2 body & 3 body system
orbit each other & then fly by



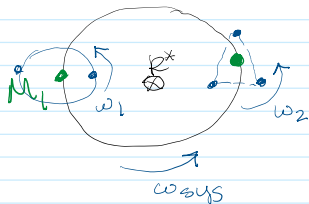
$$V = \sqrt{\frac{M_1 + M_2}{r_{12}}}$$

$$r_{12} = 2 \quad V = \sqrt{\frac{2}{5/2}} = \sqrt{\frac{4}{5}}$$

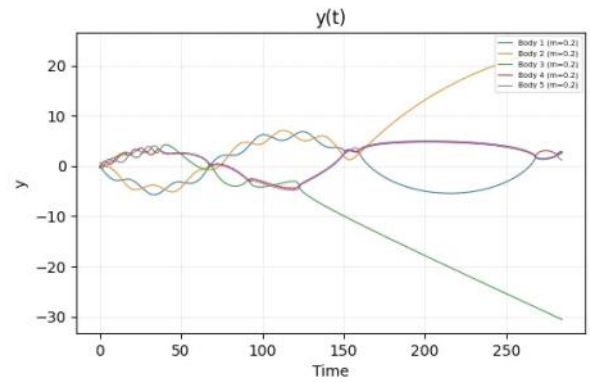
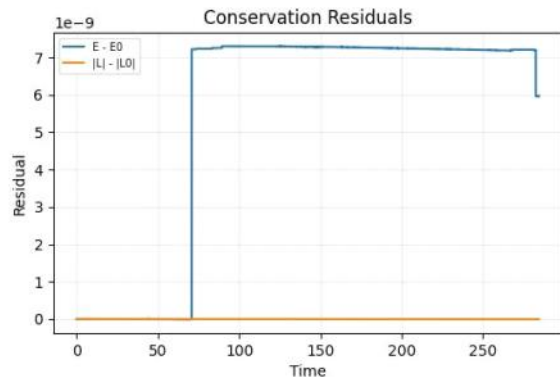
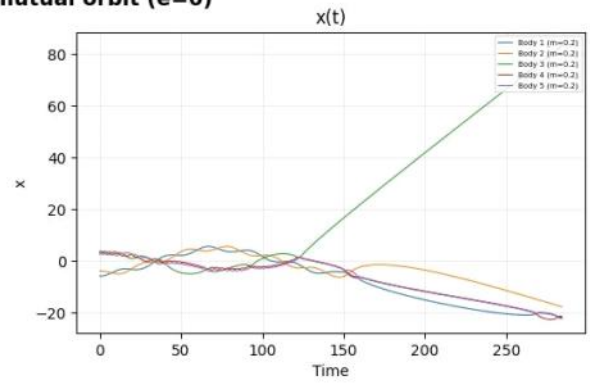
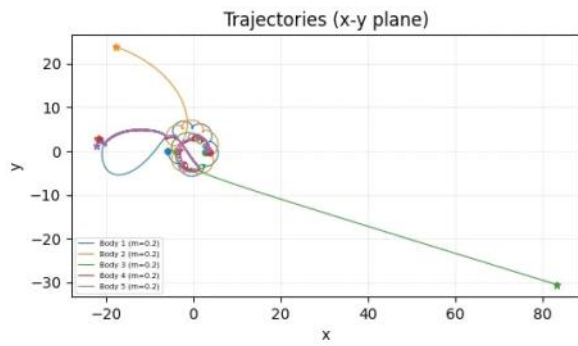
$$r_{12} = r_{23} = r_{13} = 1 \quad V = \sqrt{\frac{3}{5}}$$

start w/ all masses equal: $m = 1/5$

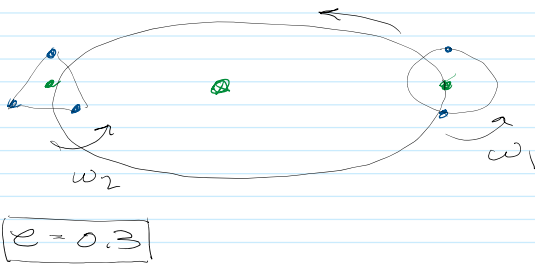
A) circular orbit around mutual COM



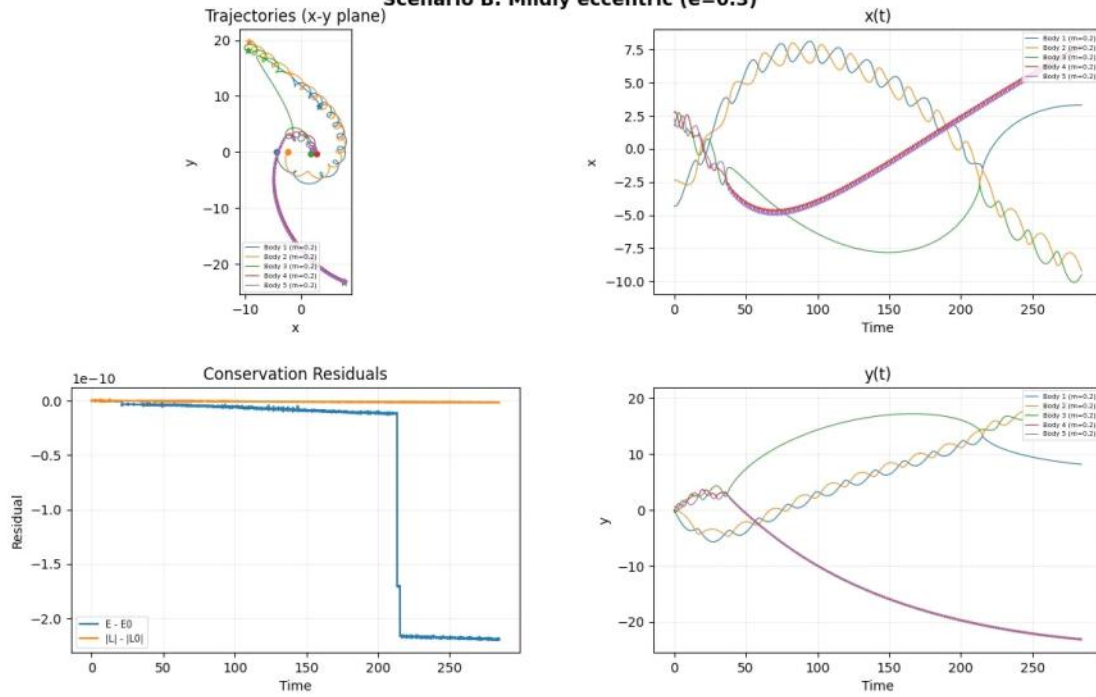
Scenario A: Circular mutual orbit ($e=0$)



B) elliptical orbit around mutual Com

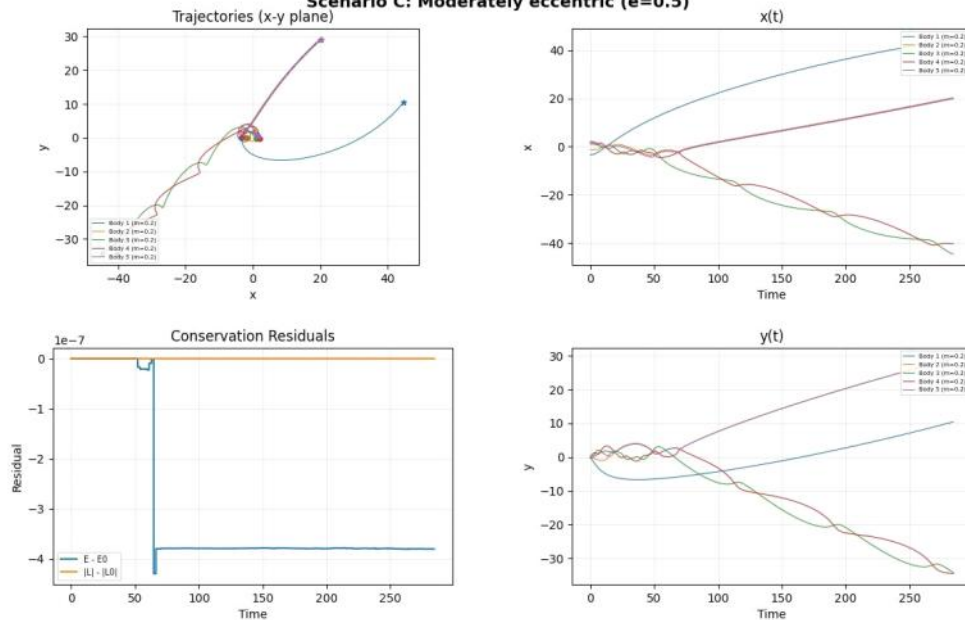


Scenario B: Mildly eccentric ($e=0.3$)



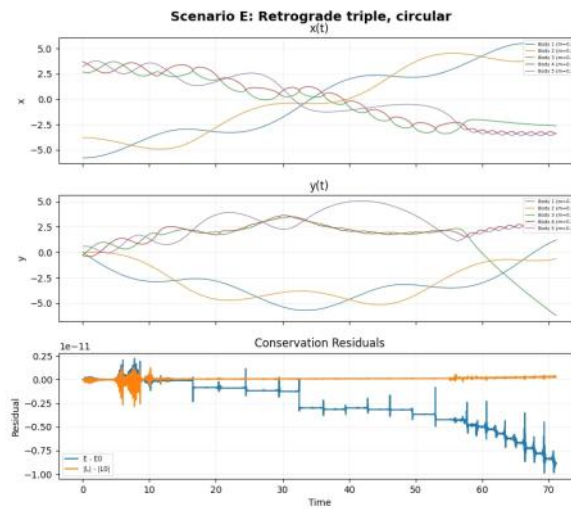
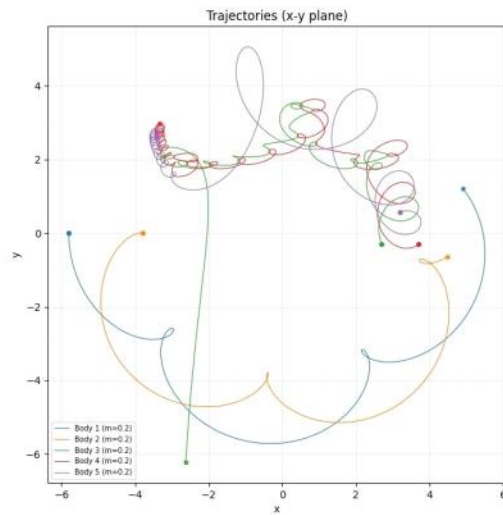
$e=0.5$

Scenario C: Moderately eccentric ($e=0.5$)



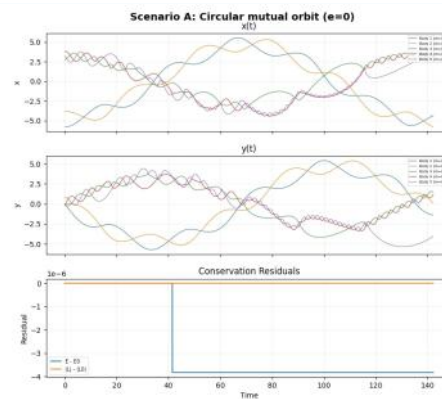
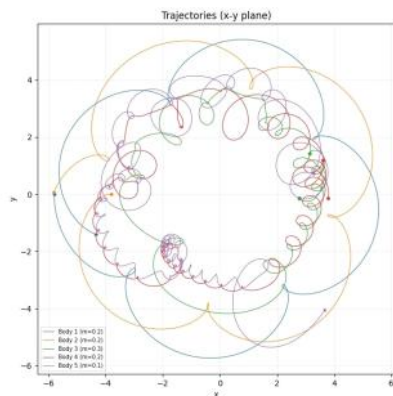
This has 3 body system rotating retrograde to 2 body system, in circular orbit

Scenario E: Retrograde triple, circular



This example shows impact of non-equal masses & their impact on the trajectories of other bodies.

Scenario A: Circular mutual orbit ($e=0$)



In all the scenarios I tested, given enough time at least one body was ejected. The motion is very chaotic & hard to predict. Lighter bodies are more impacted in their orbits when perturbed by other systems.

ASEN-6062
Comp HWK #2
Nicole James

Create restricted 4 body system
add 'massless' body to 3BP.

$\vec{R}_3$ measured from com

repeat 1a,b,c focused on 4th body
 $\vec{R}_0 = \vec{0}$

for b & c plot perturbed & unperturbed trajectory
provide EoM for body 4 & plot trajectories

CR4BP

3a) Verify conservation residuals magnitude change w/ tolerance
 $m_1 + m_2 + m_3 = 1$

$$\vec{R}_1 = \vec{R}_{12} = \vec{R}_2 - \vec{R}_1$$

$$\vec{R}_2 = \frac{1}{m_1 + m_2} [m_1(\vec{R}_3 - \vec{R}_1) + m_2(\vec{R}_3 - \vec{R}_2)]$$

$$\vec{R}_N = \vec{R}_C$$

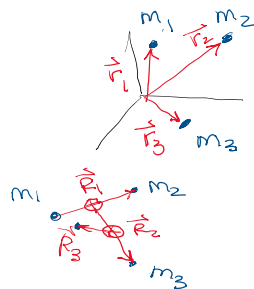
$$\vec{R}_{N-1} = \frac{1}{\sum_{i=1}^{N-1} m_i} \sum_{i=1}^{N-1} m_i \vec{R}_{iN}$$

use Euler w/ unequal mass

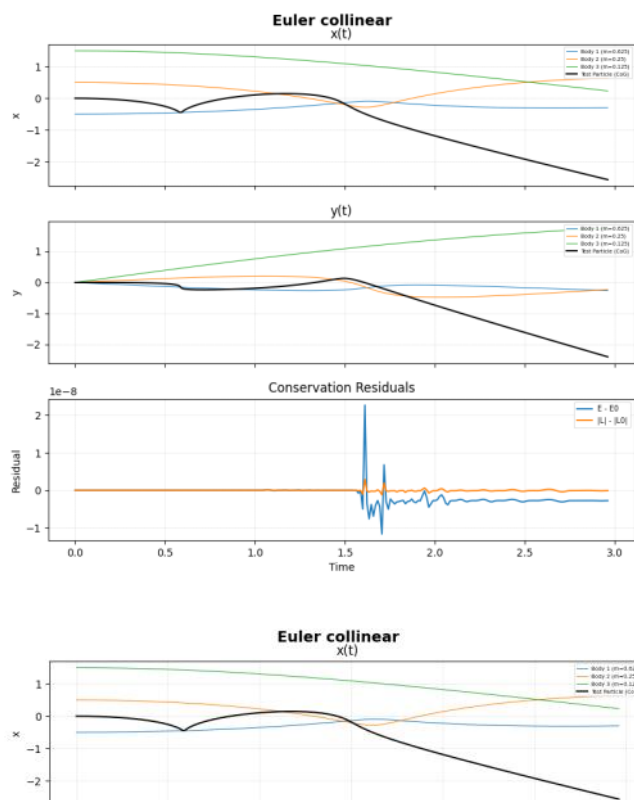
$$M = [\frac{5}{8} \quad \frac{1}{4} \quad \frac{1}{8}]$$

$$\vec{r} = [-a, 0, a]$$

com
 m_1 m_2 m_3

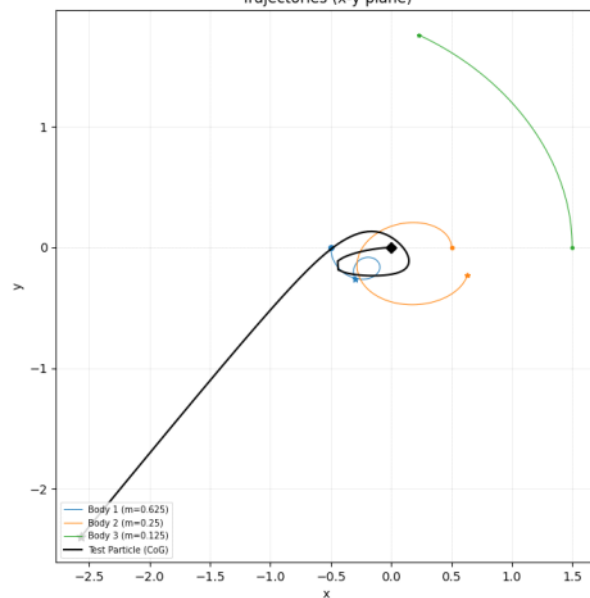


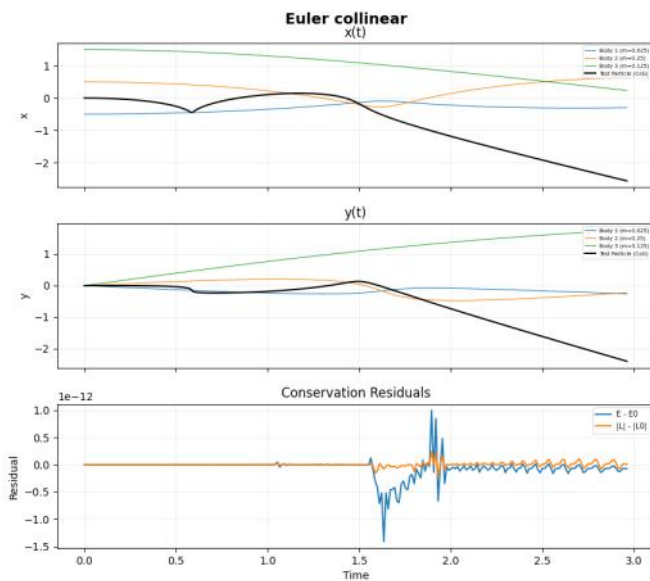
Magnitude of errors in conserved values
align w/ tolerance of integrator:



Euler collinear

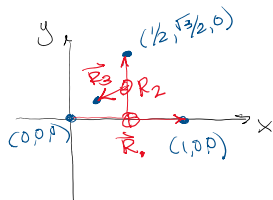
Trajectories (x-y plane)





-2.5 -2.0 -1.5 -1.0 -0.5 0.0 0.5 1.0 1.5
x

3b)



$$\mu_1 = \frac{m_1 m_2}{m_1 + m_2}$$

$$\mu_2 = \frac{(m_1 + m_2) m_3}{m_1 + m_2 + m_3}$$

$$\ddot{\vec{r}} = G \sum_{i=1}^3 \frac{m_i (\vec{r}_4 - \vec{r}_i)}{|\vec{r}_4 - \vec{r}_i|^3}$$

$$L = T - U$$

$$\mu_i \ddot{\vec{r}}_i = - \frac{\partial U}{\partial \vec{r}_i}$$

$$\vec{x} = \begin{bmatrix} \vec{r} \\ \vec{p} \end{bmatrix}$$

$$\ddot{\vec{r}} = \frac{1}{\mu_i} \left(- \frac{\partial U}{\partial \vec{r}_i} \right)$$

$$U = -G \sum_{i,j} \frac{m_i m_j}{|\vec{r}_{ij}|}$$

$$\frac{\partial U}{\partial \vec{r}_i} = -G \sum_j \frac{m_i m_j}{|\vec{r}_{ij}|^3} \vec{r}_{ij} \frac{\partial \vec{r}_{ij}}{\partial \vec{r}_i}$$

$$\frac{\partial \vec{r}_{ij}}{\partial \vec{r}_i} = \begin{cases} 0 & l > j-1 \\ 1 & l = j-1 \\ \frac{m_{i+l}}{m_{i+l-1}} & j-2 \leq l < i-1 \\ -\frac{m_{i-1}}{m_i} & l = i-1 \\ 0 & l < i-1 \end{cases}$$

$\vec{r}_3$ is system COM

$$\vec{r}_4 = \vec{r}_3$$

in COM frame, $\vec{r}_3 = \vec{0}$

$$m = \begin{bmatrix} 1/2 & 1/3 & 1/6 \end{bmatrix} \quad M = 1$$

$$\vec{r}_1 = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\vec{r}_2 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

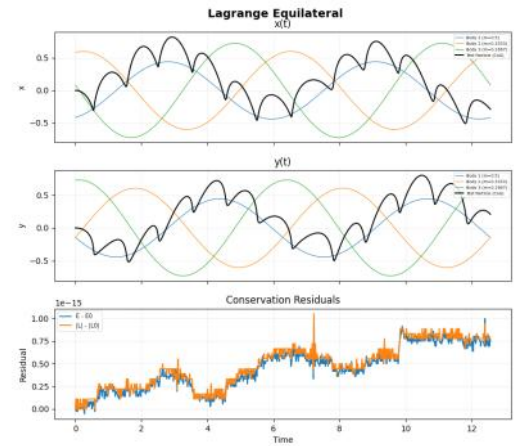
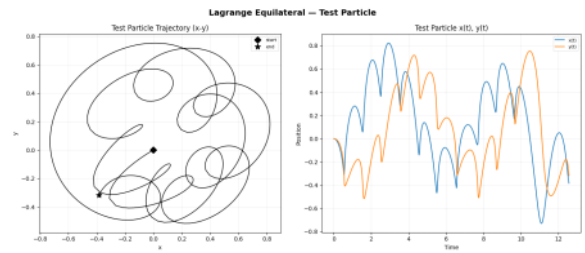
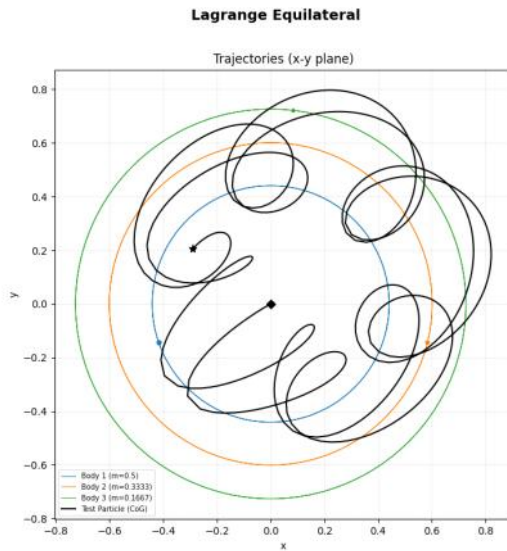
$$\vec{r}_3 = \begin{bmatrix} 1/2 & \sqrt{3}/2 & 0 \end{bmatrix}$$

$$\vec{r}_4 = \begin{bmatrix} 5/2 & \sqrt{3}/2 & 0 \end{bmatrix}$$

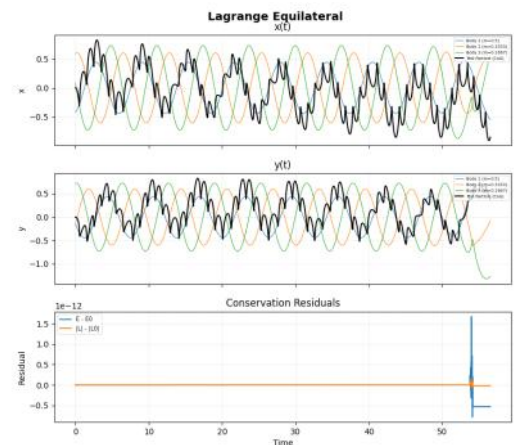
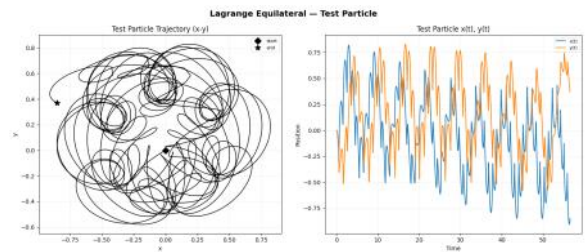
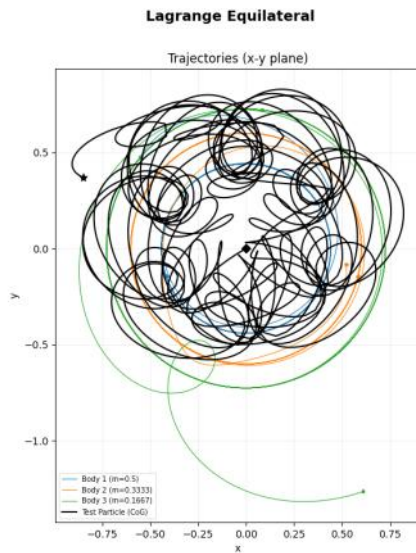
$$\dot{\vec{r}} = \omega \times \vec{r} \quad \omega = 1$$

Eqs of motion

This plot shows an unperturbed system. The fourth body stays in the system & does not escape.

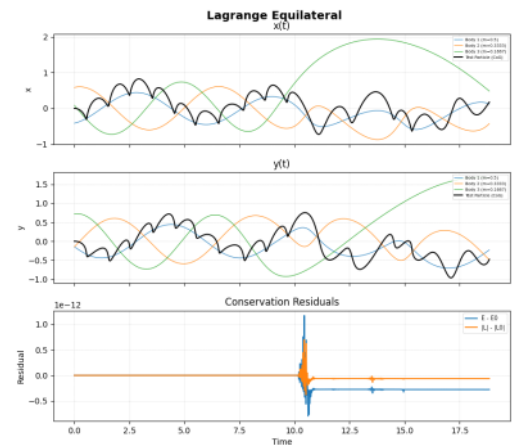
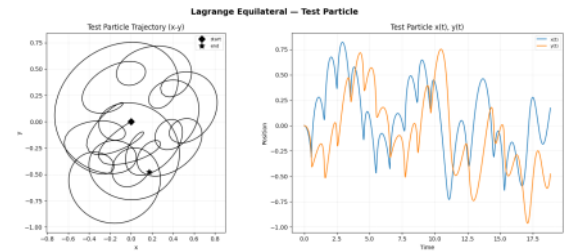
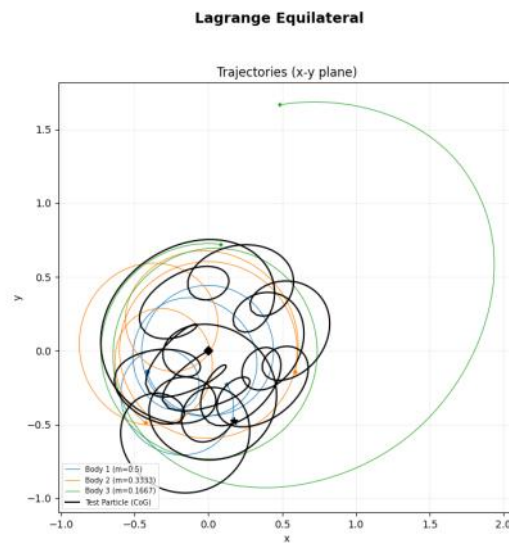


This plot shows 9 periods of the unperturbed system. The 3rd body escapes but the fourth body continues to be bound to the system.



Then a perturbation of 0.001 was introduced in the x & y position of the heaviest body. These plots show 3 periods of the orbits. The third body is now perturbed.

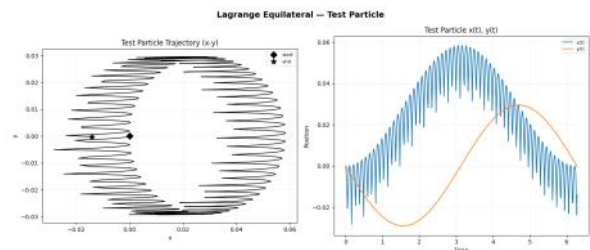
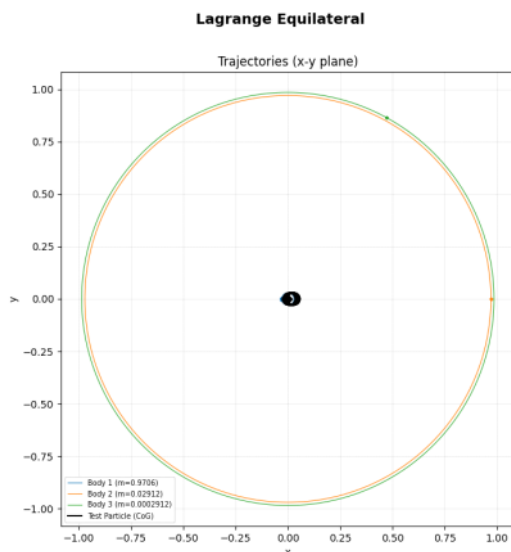
x & y position of the heaviest body. These plots show 3 periods of the orbits. The third body escapes significantly faster in the perturbed environment.



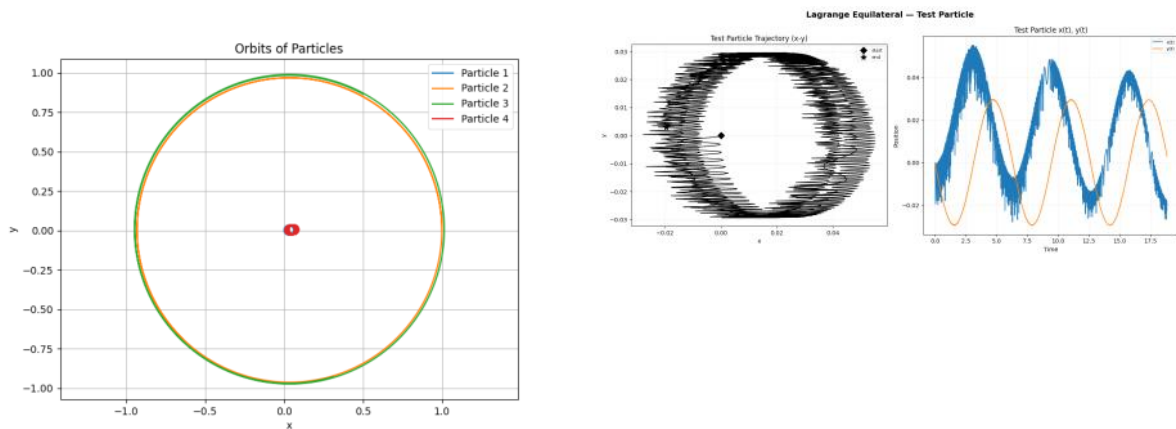
3c) Routing criteria

$$M = [1, 0.03, 0.003] / M$$

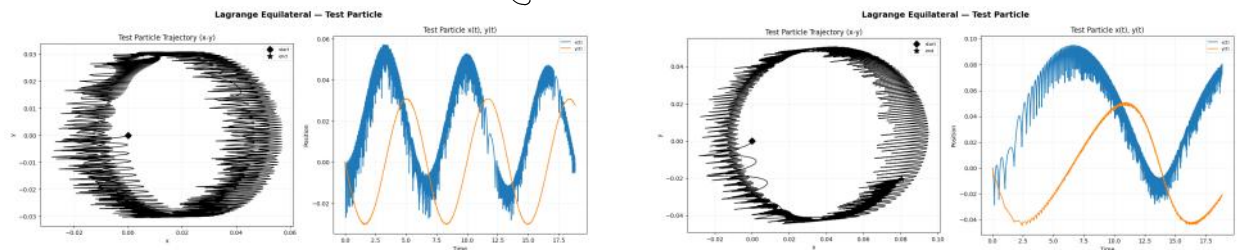
This plot shows the non-perturbed system. The fourth body stays close to the largest body.



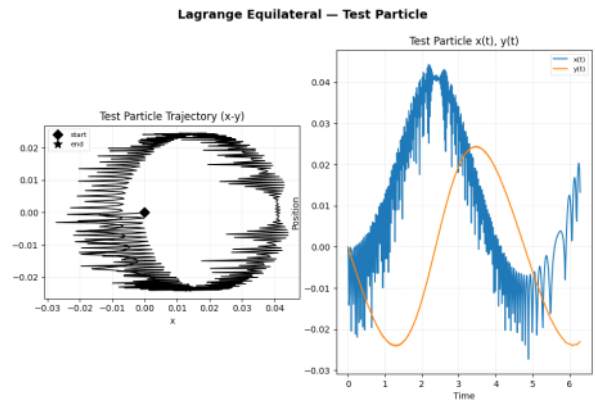
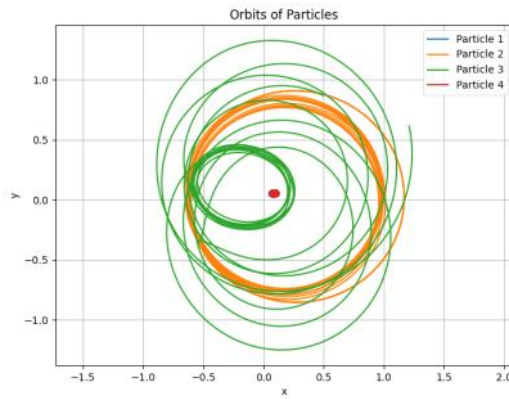
This plot shows the system by 0.001 in the x & y directions of body 1 (nearest). The perturbation is not enough to impact the stability of the orbit. The fourth body stays close to m_1 throughout the period.



These two plots show the fourth body trajectories with a perturbation of 0.01 (left) and 0.1 (right) in the x & y direction of m_1 . The body continues to stay close to m_1 .



These plots show the system perturbed by 0.5 in the x & y direction of m_1 . The third body's orbit becomes unstable but still does not escape. The fourth body continues to stay in orbit around m_1 with influences on its trajectory by the other bodies.



Adding a fourth body to these system that does not impact the underlying dynamics & plotting its trajectory shows the complex interactions & forces between the bodies. The attractive forces are complex & hard to predict. Introducing perturbations accelerates orbit instability & escape of the smallest body.

Systems that meet the Routhy criteria are stable until large perturbations are introduced. However, it takes many periods for the impact to the body trajectories to become apparent. It is possible to force a Routhy system to become unstable if the perturbation is "large".